Aetheris Five-Phase Build Prompts

Sequential prompts for building the circular materials marketplace

How to use this set
Run the phases in order against the existing Aetheris project. These prompts are a newly composed build sequence, not a verbatim transcript of earlier chat messages. The current prototype uses rules-based matching; Phase 4 describes future AI work, not a live feature.

PHASE 01

Establish the product and baseline

Mission Continue the existing Aetheris project. Do not replace it with a different product.

Aetheris is a marketplace where organizations can offer surplus materials or request resources. Its discovery graph connects surplus streams to possible recovery processes, recovered materials, and industries that may use them.

Before changing anything

- Inspect the repository, README, application files, and Git status. Identify what is implemented, what is demo data, and what is only planned.
- Preserve existing user changes and project history.
- Summarize the current architecture and user flow, identify blockers, then make a small plan based on the actual code.

Product principle Show routes as hypotheses that require validation. Never present sample companies, matches, savings, environmental outcomes, or AI results as verified facts.

PHASE 02

Build the marketplace experience

Mission Make the existing Aetheris interface work as a clear two-sided industrial materials marketplace.

User roles

A seller lists available surplus. A buyer posts a request for a material input. Processors and downstream industries may appear as route participants, but do not invent real companies or imply that they are verified.

Implement

- Browse surplus offers and buyer requests; search materials, companies, industries, and regions.
- Filter by listing type, material family, industry, region, and saved items. Sort by recency, quantity, price, or mapped routes.
- Save listings and add offers or requests with material details, quantity, region, and optional price.
- Open a listing in its related discovery pathway.

Guardrails Clearly label synthetic demo listings. If there is no backend, keep entries local and say so. Do not imply forms publish listings, contact companies, arrange delivery, or process payments.

PHASE 03

Make the discovery graph the core feature

Mission Make the graph the main way users discover where a surplus material could go.

Graph structure

Represent each possible route as: surplus stream to recovery process to recovered material to potential buyer industry. Use a shared material-family catalog and explicit route data.

- Make routes searchable and filterable by industry.
- Support network, pathway-list, and agriculture-cycle views.
- Let users select nodes and routes to inspect the process, related listings, assumptions, and qualification checks.

Agriculture cycle

Show food and crop residues going through source separation and anaerobic digestion. Biogas may be used for farm energy. Digestate is a separate stream that needs treatment, lab testing, and approval before any site-specific soil use. Connect crop residues back into collection.

Guardrails Do not invent processor capacity, conversion yield, product certification, safety approval, or economic value. Make missing checks visible.

PHASE 04

Add an evidence-grounded AI pipeline

Mission Add AI only where it improves material discovery. Keep rules-based behavior clearly distinguished from AI-generated results.

Proposed workflow

- Extract material names and stated properties from an offer, request, or uploaded specification.
- Resolve inconsistent terms to a shared material catalog.
- Retrieve relevant evidence from approved material, process, and qualification sources.
- Search the material graph for candidate direct or multi-step routes.
- Check candidates against stated properties, process needs, quantity, geography, and known limitations.
- Return a structured explanation with source references, unknowns, and required human checks.
- Let a qualified person review any route before anyone acts on it.

Safety and truthfulness

- Keep provider credentials on the server, never in browser code. Treat user documents and retrieved content as untrusted data.
- Do not let a model invent compatibility, certifications, yields, prices, or environmental impact.
- If no provider or evidence source is configured, show deterministic results and label them accurately.
- Never present a mock response as live AI.

PHASE 05

Prepare the SaaS product and handover

Mission Prepare Aetheris for a credible demonstration and outline what is needed for a real SaaS service.

Product readiness

Inspect the actual implementation first. Where supported by the architecture, add organization accounts, role-based access, persistent listings, verified material profiles, processor capacity, private sourcing requests, audit history, and opt-in match notifications. Do not simulate these features with static UI.

Quality and data

- Run the project checks and report their actual results. Check marketplace flows, map interactions, empty and error states, responsive behavior, and accessibility.
- Use synthetic data unless verified records are available. Do not claim measured waste diversion, cost savings, or emissions impact without evidence.

Deployment and handover

- Deploy only when required account access and configuration are available. Keep API keys and private business data out of the browser and repository.
- Update the README with setup, demo instructions, implemented AI status, known limitations, and remaining deployment steps.
- Separate what is implemented, tested, verified, blocked, and planned. Identify files changed and explain how to run the demo.

Final rule Do not claim the SaaS backend, live AI pipeline, or deployment is complete unless it actually is.